

An Exact Area Gate for Entropic Accounts of Geometry: Volume Laws, Count Splitting, and the Causal-Determinism Mechanism

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Entropic accounts of gravity require that the number of independent states associated with a spatial region scale with the area of its boundary. Generic hidden multiplicity scales with volume instead, so the requirement is a gate that any candidate mechanism must pass, and the expected outcome at the gate is failure. This paper reports exact measurements at that gate for declared mechanism classes on two-dimensional lattices, with every count exact and every closed form reproduced across two decades of region size. An independent-spin control shows the guaranteed volume law. The two most concrete constraint mechanisms, a Gauss-law ensemble and its gauge-quotient reading, fail the gate with an exact volume law in the primary count, the conditional entropy of the region given its boundary. The same ensemble carries an exact area law in its mutual information with the rest of the lattice, so the two counts split, and the split is measured exactly. A third arm shows that an observer restricted to boundary data obtains an area-scaling image while the hidden interior keeps a volume ambiguity, so an access limit is not a distinguishability limit. An elementary lemma makes the negatives structural. Any translation-invariant ensemble with positive entropy density forces a volume law in the primary count, so every mechanism class in that regime fails before it is built. The lemma points to ensembles with zero entropy density, whose natural uninserted origin is deterministic local dynamics with the region read as a spacetime region, and a causal-cone theorem shows that every deterministic nearest-neighbor rule bounds the states of a square spacetime patch by its perimeter, with nothing tuned. Nothing in this paper bears on physical gravity.

I. INTRODUCTION

Every derivation of gravity from entropy begins by assuming that the entropy associated with a region of space is proportional to the area of the region's boundary. Jacobson derived the Einstein equation as an equation of state from exactly that proportionality together with a local form of the Clausius relation [1]. The recent lineage descending from that argument, an entropic action built from quantum relative entropy [2], a modular-theory refinement on bifurcate Killing horizons [3], and an equilibrium treatment of stretched light cones [4], refines the machinery but keeps the same input. In each case the proportionality of entropy to area is assumed, not derived. None of these works says what the microscopic states are, or why counting them should follow the boundary of a region rather than its bulk.

That assumption is where a constructive program must start, because it is where generic counting fails. If an observer sees a projection of a hidden system, the hidden states compatible with a region of observations generically multiply with the number of hidden degrees of freedom inside the region, which grows with the region's volume. A framework in which gravity is to emerge from hidden-state counting therefore faces a gate. Some declared structure must make the independently distinguishable states of a region scale with its boundary, and must do so for a structural reason that was not itself inserted. The present campaign calls this the area gate, treats failure as the expected outcome, and treats a clean negative as a result.

This paper reports the campaign's passage to and through that gate at the level of exact counts. Four results are reported, and each carries an explicit evidence label. First, an instrument layer, validated against closed forms, that measures conditional entropy and relative entropy of hidden-state ensembles at finite resolution and is invariant under relabeling of the hidden coordinates. Second, exact measurements of the gate itself for three declared arms on two-dimensional lattices, an independent-spin control, a Gauss-law constraint ensemble read two ways, and a boundary-limited observer, across two decades of region size with every count an integer computed in closed form and verified at every size. The constraint mechanisms fail the gate in the primary count, and the failure is exact. The same measurement finds an exact area law in a secondary count, the mutual information between the region and the rest of the lattice, so the two counts split and the split itself is a measured, exact structure. Third, an elementary lemma, proved here, showing that any translation-invariant ensemble with positive entropy density must fail the gate, which converts the measured negatives from instances into a classification. Fourth, a theorem about deterministic local dynamics, proved here, showing that the states of a square spacetime patch under any deterministic nearest-neighbor rule are bounded by causal-cone data whose size scales with the patch perimeter. This identifies a mechanism class, causal determinism, that satisfies the gate's scaling requirement for every rule in the class, with nothing tuned. A measured instance and its structural-origin audit are specified, and their numbers are bound to a committed evidence record.

The scope of these claims is narrow. The lemma and the theorem are statements about lattice ensembles and

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cellular automata. The measurements are exact counts in declared toy constructions. No claim is made about physical spacetime, about real gravitation, or about the correctness of any entropic-gravity proposal. What the paper establishes is a discipline. It says which mechanism classes cannot produce the scaling those proposals assume, it exhibits one class that provably can, and it shows that the choice of operational count decides the verdict, so that any future claim of area scaling must state its count before it states its result.

II. THE GATE AND THE OPERATIONAL COUNT

The setting is the campaign’s standard one. A hidden state Z lives in a declared state space, an observer sees $X = \pi(Z)$ for a declared observation map π , and every ensemble, resolution, and measure is fixed before any claim-bearing run. The experiments of this paper use finite lattices, so every entropy below is a finite Shannon entropy in bits and every count is exact.

Fix a region Q_R of linear size R , with boundary ∂Q_R and interior Q_R° . The gate asks how the number of independently distinguishable hidden states associated with Q_R scales with R . On the two-dimensional lattices used here, volume scaling means growth proportional to R^2 and area scaling means growth proportional to R . The phrase “independently distinguishable states associated with a region” admits several formalizations, and they disagree, so the choice had to be made before any run. The track document records the decision in these words. “The primary count is the conditional entropy of the region’s hidden state at resolution delta given the boundary data, $H(\text{region} | \text{boundary})$, because it directly counts interior distinctions not fixed by the boundary, which is what the gate’s independence requirement is about.” The mutual information $I(\text{region}; \text{complement})$ is a mandatory secondary witness, computed in every arm, because the two counts can diverge and the divergence is itself informative. Any arm whose verdict differs between the two counts is reported under both, with the primary count deciding the gate.

One standing warning from the track design governs the reading of every arm. Concentration of measure near a boundary is not area scaling. A measure concentrated in a thin shell near the boundary still counts volume-independent distinctions unless the interior distinctions are genuinely not independent. The gate is about independence of distinctions, not about where the measure sits, and the third arm below is that warning made quantitative.

III. THE INSTRUMENT LAYER

Before the gate could be measured, the instruments had to be. The instrument experiment extends the campaign’s

projection-entropy stack [5, 6] to the full conditional-entropy surface and to relative entropy between deformed and undeformed fiber measures, and validates every part against exact closed forms. All numbers in this section are from the committed evidence record of that run.

The first part reproduces the campaign’s published closed-form controls. The fold’s branch entropy is measured at one bit within 1.4×10^{-14} , the symmetric slice of the double fold at exactly 1.5 bits, the degenerate cubic at exactly zero bits, and the binned caustic entropy approaches its differential limit within 1.4×10^{-3} at bin width 10^{-5} , matching the published resolution law. The second part measures the conditional-entropy surface $H(Z_\delta | X_\epsilon)$ at hidden resolution δ and observed resolution ϵ . An aligned linear control reproduces its exact value of $\log_2 100$ bits within 1.6×10^{-14} , and both resolution sweeps converge, with successive deviations shrinking by a factor of 6.2 per refinement step in δ and 1.4 per step in ϵ . The third part measures relative entropy between fiber measures. Nested uniform fibers reproduce their exact value of 2 bits within 4.4×10^{-16} . A Gaussian pair reproduces the closed-form divergence of 0.2644 bits within 1.75×10^{-10} , and a widening control identifies that floor as range truncation, since the error falls to 1.7×10^{-16} when the integration range is widened to fourteen standard deviations. A piecewise-deformed fold matches the binary divergence closed form within 3.3×10^{-16} in every one of its 90 interior bins. The fourth part verifies reparametrization invariance. Relabeling the hidden coordinate changes no witness beyond 1.4×10^{-16} , and Jacobian-weighted re-gridding changes none beyond 8.2×10^{-16} , shrinking to 4.4×10^{-16} at quarter resolution, which identifies the residual as quadrature rounding rather than a coordinate dependence. An instrument whose entropies moved under a hidden coordinate change would be measuring the coordinates, not the ensemble, and this check is what entitles the counts below to be called properties of the constructions.

IV. THREE ARMS AT THE AREA GATE

The gate experiment measures three declared arms on two-dimensional lattices at region sizes R from 3 to 301, fifteen sizes spanning two decades. Each arm reports the primary count $H(Q_R^\circ | \partial Q_R)$ and the secondary count $I(Q_R; \text{complement})$ at every size. Every reported number is an exact integer count in bits, every closed form quoted below is reproduced by the measured count at every one of the fifteen sizes, and the scaling fits compare a free power law against the two fixed alternatives, exponent 2 for volume and exponent 1 for area, all declared before the run. All numbers in this section are from the committed evidence record of the gate run.

A. Arm A, independent spins

The control arm places an independent uniform binary spin on every site. The primary count is exactly $(R - 2)^2$ bits at every R , one bit for every interior site, and the mutual information with the complement is exactly zero at every R . The fitted free slope over the full range is 2.328, and the fixed-exponent comparison prefers volume, with residual 0.869 against exponent 2 versus 2.862 against exponent 1. The free slope exceeds 2 because the offset in $(R - 2)^2$ bends the small end of the log-log line, not because the count is superextensive. This is the guaranteed volume law that the gate exists to beat.

B. Arm B, Gauss-law constraint and gauge quotient

The first serious mechanisms are interior constraint and gauge redundancy. Both are realized in one construction. Binary variables live on the edges of a periodic square lattice, a torus of side $L = 2R + 9$ so that the embedding lattice grows with the region and the padding stays fixed, and the ensemble is uniform over the configurations satisfying a constraint of Gauss-law type at every vertex, meaning that every vertex touches an even number of occupied edges. The satisfying configurations form the cycle space of the lattice graph, the set of edge subsets in which every vertex has even degree, which is a linear space over the two-element field. The Gauss-law reading counts this constrained ensemble directly. The gauge-quotient reading counts equivalence classes under a symmetry group acting on vertex variables, and its quotient count is the same cycle-space count, so the two mechanism classes share one measurement. Every entropy of this ensemble equals the dimension of a cycle space, and those dimensions are computed exactly by counting connected components of the relevant graphs, so no estimator appears anywhere in the arm.

The region Q_R is an $R \times R$ patch of the torus. The measured primary count is exactly $(R - 1)^2$ bits at every R . Given the entire boundary of the patch, the interior still carries one exact bit per interior plaquette. The fitted free slope is 2.132, with residual 0.333 against exponent 2 versus 2.405 against exponent 1, so the arm is classified volume and the two most concrete constraint mechanisms fail the gate in the primary count. The constraint removes exactly the boundary-linked distinctions and not one more. The interior of a Gauss-law ensemble is not boundary data in disguise. It is an extensive reservoir of independent distinctions, and the gate's independence requirement sees it.

The same ensemble then produces the finding beyond the negative. Its mutual information with the complement is exactly $4R - 5$ bits at every R , an exact area law, with fitted free slope 1.086 and residual 0.220 against exponent 1 versus 1.939 against exponent 2. The marginal entropy of the patch equals $(R - 1)^2 + 4R - 5$ bits exactly, the two pieces adding with no interference term. The counts

TABLE I. The five measured series of the gate run. Each closed form is reproduced exactly by the measured count at all fifteen region sizes. The free slope is the fitted exponent of a free power law over the full range, and the classification compares residuals against the fixed alternatives, exponent 2 for volume and exponent 1 for area, declared before the run.

Series	Closed form	Free slope	Class
arm A primary	$(R - 2)^2$	2.328	volume
arm B primary	$(R - 1)^2$	2.132	volume
arm B mutual information	$4R - 5$	1.086	area
arm C access image	$4R - 4$	1.066	area
arm C hidden residual	$(R - 2)^2$	2.328	volume

split. Constraint structure does put area scaling into the correlations between a region and its complement, exactly where the classical Gibbs-ensemble confound anticipated by the track design would put it, while leaving interior independence extensive. The preregistered decision quoted in Sec. II is what makes this a clean verdict rather than an ambiguity. An entropic-gravity reading built on this mechanism class would have to live on the correlation count, which the decision relegated to secondary, and would have to justify that choice against the independence requirement the gate is about. The split is measured in one construction on two-dimensional lattices, and nothing here decides how general it is.

C. Arm C, boundary-limited access

The third arm implements the mechanism most tempting to claim and measures why it does not count. The observation map reads the interior of the region only through its boundary data, so the observer-accessible image carries exactly $4R - 4$ bits by construction, area scaling with fitted free slope 1.066. The hidden interior, however, retains a residual of exactly $(R - 2)^2$ bits that no observation can reach, a volume law with fitted free slope 2.328, identical to the control arm. Restricting access manufactures an area-scaling image without removing a single interior distinction. The distinctions are still there, still independent, and still extensive. This is the standing warning of the track design made quantitative. An access limit is not a distinguishability limit, and an area law obtained by refusing to look at the interior is not a pass at this gate.

Figure 1 shows all five series and Table I registers their closed forms, fitted slopes, and classifications. The verdict of the gate run is a negative in the primary count for every mechanism class measured, together with an exact measured split between the independence count and the correlation count in the constraint arm. The remaining mechanism classes named in the track design, error-correcting encodings with a structural rather than inserted origin, subextensive long-range correlation, and projection rank constraints, were not constructed in this

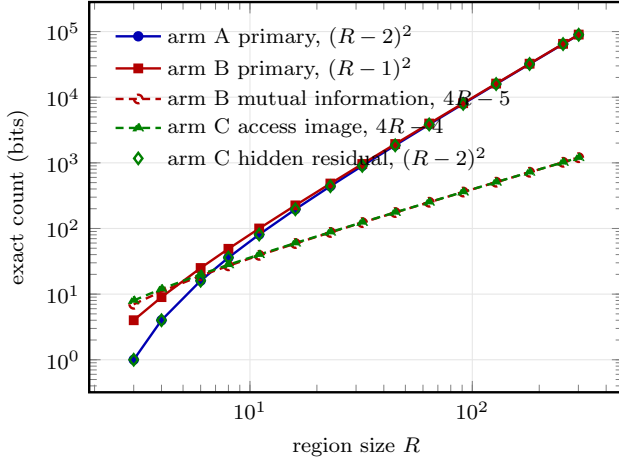


FIG. 1. The five exact counts of the area-gate run versus region size, on logarithmic axes. The series separate into a slope-two family, the volume laws of the independent-spin control, the Gauss-law primary count, and the hidden residual of the boundary-limited arm, and a slope-one family, the Gauss-law mutual information and the access image of the boundary-limited arm. The hidden residual of arm C coincides exactly with the arm A control because both count $(R-2)^2$ interior bits, so its open diamonds sit on the control line. Every point is an exact integer count read from the committed evidence record.

run, and the gate stays open for them.

V. THE ENTROPY-DENSITY LEMMA

The measured negatives are not accidents of the particular arms. They are forced by an elementary property that all three arms share, a positive density of entropy per site. The following lemma makes this exact and is the paper's organizing statement.

Lemma 1 (Positive entropy density forces a volume law). *Let each site of the two-dimensional integer lattice carry a symbol from a finite alphabet A , let the ensemble of configurations be invariant under lattice translations, let Q_R denote the square of $R \times R$ sites with boundary ∂Q_R and interior Q_R° , and let*

$$h = \inf_{R \geq 1} \frac{H(Q_R)}{R^2} \quad (1)$$

be the entropy density in bits per site. Then for every $R \geq 2$,

$$H(Q_R^\circ | \partial Q_R) \geq hR^2 - cR, \quad c = 4 \log_2 |A|. \quad (2)$$

Consequently, if the primary count is bounded by aR for all R and some constant a , then $h = 0$.

Proof. Translation invariance and subadditivity of joint entropy give $H(Q_{mR}) \leq m^2 H(Q_R)$ for all m , since Q_{mR}

is covered by m^2 translates of Q_R whose marginal entropies are equal, and joint entropy is at most the sum of marginals [7]. By the standard subadditivity argument the sequence $H(Q_R)/R^2$ converges and its limit equals the stated infimum, so $H(Q_R) \geq hR^2$ for every R . The chain rule splits the region as

$$H(Q_R) = H(\partial Q_R) + H(Q_R^\circ | \partial Q_R). \quad (3)$$

The boundary has $4R-4$ sites, each carrying at most $\log_2 |A|$ bits, so $H(\partial Q_R) \leq (4R-4) \log_2 |A| \leq cR$, and (2) follows. If additionally $H(Q_R^\circ | \partial Q_R) \leq aR$ for all R , then $h \leq (a+c)/R$ for every R , so $h = 0$. $\square$

The lemma converts the gate run into a classification. Arm A has entropy density one bit per site. Arm B has positive entropy density, one bit per plaquette in the limit, which is why its constraint could remove only a perimeter's worth of distinctions. Arm C inherits the density of its hidden ensemble untouched, because access does not enter the lemma at all. Every mechanism class whose declared ensemble has positive entropy density fails the primary count before it is constructed, and this includes generic realizations of most of the mechanism families the track design listed. The lemma is proved for translation-invariant ensembles on two-dimensional lattices with finite alphabets, and the same argument runs in any dimension with volume and boundary read accordingly. A mechanism can pass the gate only from inside the class the lemma leaves open, ensembles whose bulk entropy density is exactly zero.

VI. THE CAUSAL-DETERMINISM MECHANISM

A. From zero density to determinism

Zero entropy density is a strange-looking requirement for an ensemble that must still contain states worth counting, but one class satisfies it without any insertion. If the hidden configuration is generated by deterministic local dynamics from initial data, and the region is read as a spacetime region, a patch of the history rather than a snapshot, then all randomness lives on the initial surface and the bulk of the patch adds none. The track recorded this as a corollary decision before any run of the mechanism. For deterministic dynamics the conditional count $H(\text{interior} | \text{boundary})$ is exactly zero, which is degenerate rather than informative, so the operational count of states associated with a spacetime region is the marginal entropy of the region, compared against spacetime volume and spacetime boundary, with the conditional count still computed and reported. A pass additionally requires a structural-origin audit. The same construction with the structural feature removed, by adding bulk noise, must revert to a volume law, showing that the scaling tracks the declared structure and nothing tuned.

B. The cone bound

The mechanism's scaling is not a property of a favorable rule. It is a theorem about the entire class.

Theorem 1 (Causal-cone bound). *Let a deterministic rule of radius one update a row of binary cells on the one-dimensional integer lattice, each cell's next value a fixed function of its own value and the values of its two nearest neighbors. Read the resulting spacetime as a plane whose rows are successive time steps, and let P_R be a patch of R consecutive cells over R consecutive time steps. Then the configuration of P_R is a function of the initial data on a segment of $3R - 2$ cells, and therefore for every probability ensemble on initial conditions*

$$H(P_R) \leq 3R - 2 \text{ bits.} \quad (4)$$

Proof. Let the patch occupy columns x_0 through $x_0 + R - 1$ and rows t_0 through $t_0 + R - 1$, and let S be the segment of row t_0 from column $x_0 - (R - 1)$ to column $x_0 + R - 1 + (R - 1)$, which has $3R - 2$ cells. We show by induction on k that the restriction of row $t_0 + k$ to columns $x_0 - (R - 1 - k)$ through $x_0 + R - 1 + (R - 1 - k)$ is a function of S . For $k = 0$ this restriction is a subset of S . For the step, each cell of row $t_0 + k + 1$ in the stated range depends only on three cells of row $t_0 + k$ lying in the previous range, since the rule has radius one. Taking $k = 0$ through $R - 1$ shows every cell of the patch is a function of S . The entropy of a function of a random variable is at most the entropy of that variable, which is at most its cell count in bits, giving (4). $\square$

The bound $3R - 2$ is linear in R , the scaling of the patch perimeter, while the patch has R^2 cells. The bound holds for any rule of the class and any initial ensemble, so nothing about it is tuned, selected, or fitted, and for an alphabet of q symbols the same argument gives $(3R - 2) \log_2 q$. For a rule of radius r the segment has $R + 2r(R - 1)$ cells, still perimeter scaling for every fixed r . The companion fact recorded in the corollary decision is equally elementary.

Proposition 1 (Deterministic interiors are boundary functions). *For any deterministic radius-one rule, the interior of a spacetime patch P_R is a function of the boundary of P_R , so $H(P_R^o \mid \partial P_R) = 0$ exactly.*

Proof. The boundary contains the patch's top row and its two side columns. Given the full top row of the patch, the rule determines the next row's interior cells, whose missing endpoints are supplied by the side columns, so the next row is known in full. Iterating down the patch determines every interior cell. $\square$

The proposition is why the primary count of Sec. II is degenerate here and the marginal count decides, exactly as the corollary decision required. The causal-determinism mechanism thus passes the gate's scaling requirement in the only count that is informative for it, by a proof that quantifies over the whole rule class.

C. The measured instance and its audit

The measured companion to the theorem, the mechanism run of the campaign, instantiates the class with the elementary rule numbered ninety, the linear rule in which each cell updates to the sum modulo two of its two neighbors. For a linear rule acting on uniform initial data the patch ensemble is uniform over a linear subspace, so its entropy in bits equals the rank of that subspace over the two-element field and is computed exactly. The run measures patch counts over a sweep of patch sizes up to $R = 301$, together with the structural-origin audit, the same construction with one fresh uniform bit added to each bulk cell after its deterministic update at each step, equivalent to independent flips with probability $1/2$. The audit arm restores a positive bulk entropy density, so by Lemma 1 its count must revert toward volume if the deterministic structure, and nothing else, was carrying the perimeter scaling.

The numbers of this subsection are bound to the committed evidence record of the mechanism run. At the largest patch size the rule-ninety count is 901 bits against the cone bound of Eq. (4), a ratio of 1, and the same saturation holds exactly at every measured size. Over the full sweep the deterministic arm has fitted free slope 1.042 and is classified area against the fixed alternatives. The audit arm reaches 2025 bits at its largest audited size, forty-five, exactly its cell count, with fitted free slope 2.0000, and is classified volume. The theorem, not these numbers, carries the mechanism's scaling claim. The measurement locates one rule inside its bound and the audit ties the scaling to the declared structure.

VII. RELATION TO ENTROPIC ACCOUNTS OF GEOMETRY

The results bear on the entropic-gravity lineage at exactly one point, the input every member of the lineage shares. Jacobson's derivation consumes an entropy proportional to horizon area [1]. Bianconi's entropic action postulates a relative-entropy functional whose weak-coupling limit is Einstein gravity, with no microscopic ensemble whose coarse-graining produces its density matrices [2]. Dorau and Much prove a flux-entropy equality by modular theory but still consume the entropy-area formula as an input [3]. The stretched light-cone treatment consumes the same proportionality together with an assumed temperature [4]. In every case the area law is purchased, not earned. The present paper prices that purchase in a setting where everything can be counted. Constraint and gauge structure, the mechanisms closest to how the lineage's states are usually imagined, provably do not deliver an area law in the independence count, and deliver an exact one in the correlation count, so which count a proposal's entropy refers to is not a technicality but the whole question. A proposal whose microscopic states are ever specified will face this fork, and the causal-determinism

class is, so far, the one measured mechanism family that passes the scaling requirement with a structural origin.

Two disciplines from the track’s contract govern how far such statements may be pushed, and they are the reason this paper’s claims stop where they do. The first is the reverse-engineering caveat. For any potential one can define an entropy whose gradient reproduces the corresponding force, so an entropic account earns content only when the entropy is computed from an independently specified ensemble and coarse-graining, with the target law nowhere inserted. The second is the anti-circularity contract that binds every run reported here. The ensembles, projections, measures, counts, and fit alternatives were declared before the runs, the forbidden-input list excludes any inserted area proportionality or target law, and a construction that obtained its scaling by insertion would void the run rather than pass the gate. The boundary-limited arm exists precisely to demonstrate the nearest temptation, an area law that is an artifact of access rather than a property of the ensemble, and the audit arm of the mechanism run exists to catch the corresponding temptation for determinism. Nothing here supports or undermines any entropic account of real gravity. What is established is which mechanism classes could never have supplied the states such accounts assume, and which class provably could, in models where every bit is countable.

VIII. WHAT THIS PAPER ESTABLISHES

Table II registers the claims with their labels. Read together they answer the gate question for the measured classes. Interior constraint and gauge redundancy cannot supply boundary-scaling state counts, access restriction only pretends to, positive entropy density of any origin is

disqualifying, and deterministic local dynamics read over spacetime regions is a mechanism class whose perimeter scaling is a theorem rather than a tuning. The gate remains open for the unconstructed classes named in the track design, and the campaign continues only if some declared mechanism passes the primary count without insertion.

IX. METHODS AND REPRODUCIBILITY

All instruments derive from the campaign’s sealed tolerance and instrument freezes, which fix numerical bars in advance and void themselves if edited, and the projection-entropy stack extends the published fold instruments [5] and the completed entropy track [6]. The operational-count decision and its dynamical corollary were recorded in the track document before the corresponding runs, and the forbidden-input list of the track’s anti-circularity contract applies to every construction reported here. Every result-bearing run writes an append-only JSON evidence record containing the complete declared parameter set, outcome arrays, and a canonical SHA-256 hash of the record body. The records for this paper are `eg0-instrument.json` and `eg1-area-gate.json` in the campaign repository, both carrying the exploratory label of the campaign’s evidence-grading scheme, and the mechanism run binds to `eg1b-mechanism.json`. The instrument layer is unsealed pending the campaign’s cross-substrate replication clause, and its Python implementation is the reference. Every number and every figure coordinate in this paper is read from those committed records or is an exact closed form derived in the text, with the figure table generated deterministically by a committed script with no free parameters. A reducer may summarize records but may not rewrite them.

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- [1] T. Jacobson, Thermodynamics of spacetime: the Einstein equation of state, *Phys. Rev. Lett.* **75**, 1260 (1995), arXiv:gr-qc/9504004.
 - [2] G. Bianconi, Gravity from entropy, *Phys. Rev. D* **111**, 066001 (2025), arXiv:2408.14391.
 - [3] P. Dorau and A. Much, From quantum relative entropy to the semiclassical Einstein equations, *Phys. Rev. Lett.* **136**, 091602 (2026), arXiv:2510.24491.
 - [4] A. Alonso-Serrano, L. J. Garay, M. Liska, and C. Lopez Pinerós, Gravity from equilibrium thermodynamics of stretched light cones, *Phys. Rev. D* **112**, 124026 (2025), arXiv:2509.08566.
 - [5] A. H. Bond, Entropy across singular projections: fiber multiplicity, caustics, and observer-relative information, Zenodo (2026), doi:10.5281/zenodo.21789011.
 - [6] A. H. Bond, Three sources of observed entropy in reversible hidden dynamics: mixing, projection multiplicity, and inaccessible degrees of freedom, companion paper of this campaign (2026).
 - [7] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. (Wiley-Interscience, Hoboken, 2006).

TABLE II. Claim register for the area-gate stage of the entropic-geometry track. Proved means a theorem with a written proof in this paper. Demonstrated in model means an exact measurement in a declared construction with append-only evidence records and preregistered alternatives. The mechanism-run row's numbers are bound to the committed evidence record and its label applies to the measurement, not to the theorems that motivate it.

Result	Claim	Label
Entropy-density lemma	positive bulk entropy density forces a volume law in the region-given-boundary count, so every mechanism class in that regime fails the gate before construction	proved
Causal-cone bound	every deterministic radius-one rule bounds the states of a square spacetime patch by cone data of perimeter size, with nothing tuned	proved
Deterministic interiors	for such rules the interior of a spacetime patch is a function of its boundary, so the conditional count is exactly zero	proved
Instrument layer	conditional-entropy and relative-entropy instruments reproduce every closed-form control and are invariant under hidden reparametrization	demonstrated in model
Gate, arm A	independent spins count exactly one bit per interior site with zero mutual information, the guaranteed volume law	demonstrated in model
Gate, arm B	Gauss-law and gauge-quotient mechanisms fail the primary count with the exact volume law $(R - 1)^2$ while carrying the exact area-law mutual information $4R - 5$, so the counts split	demonstrated in model
Gate, arm C	boundary-limited access yields an exact area-scaling image over an exact volume-law hidden residual, so an access limit is not a distinguishability limit	demonstrated in model
Mechanism run	rule-ninety patch counts against the cone bound with a bulk-noise structural-origin audit, numbers bound to the committed record	demonstrated in model
Scope	nothing in this paper bears on physical gravity, and no physics claim is made	non-claim